NAME

tkill, tgkill − send a signal to a thread

LIBRARY

Standard C library (*libc*, *−lc*)

SYNOPSIS

#include <signal.h> /* Definition of **SIG*** constants */
#include <sys/syscall.h> /* Definition of **SYS_*** constants */
#include <unistd.h>

[[deprecated]] int syscall(SYS_tkill, pid_t *tid***, int** *sig***);**

#include <signal.h>

int tgkill(pid_t *tgid***, pid_t** *tid***, int** *sig***);**

Note: glibc provides no wrapper for **tkill**(), necessitating the use of **syscall**(2).

DESCRIPTION

tgkill() sends the signal *sig* to the thread with the thread ID *tid* in the thread group *tgid*. (By contrast, **kill**(2) can be used to send a signal only to a process (i.e., thread group) as a whole, and the signal will be delivered to an arbitrary thread within that process.)

tkill() is an obsolete predecessor to **tgkill**(). It allows only the target thread ID to be specified, which may result in the wrong thread being signaled if a thread terminates and its thread ID is recycled. Avoid using this system call.

These are the raw system call interfaces, meant for internal thread library use.

RETURN VALUE

On success, zero is returned. On error, −1 is returned, and *errno* is set to indicate the error.

ERRORS

EAGAIN
: The **RLIMIT_SIGPENDING** resource limit was reached and *sig* is a real-time signal.

EAGAIN
: Insufficient kernel memory was available and *sig* is a real-time signal.

EINVAL
: An invalid thread ID, thread group ID, or signal was specified.

EPERM
: Permission denied. For the required permissions, see **kill**(2).

ESRCH
: No process with the specified thread ID (and thread group ID) exists.

STANDARDS

Linux.

HISTORY

tkill() Linux 2.4.19 / 2.5.4.

tgkill() Linux 2.5.75, glibc 2.30.

NOTES

See the description of **CLONE_THREAD** in **clone**(2) for an explanation of thread groups.

SEE ALSO

clone(2), **gettid**(2), **kill**(2), **rt_sigqueueinfo**(2)